



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

PROJECT TITLE- INTERVIEW AI

The domain of the Project:
Generative AI

Team Mentors (and their designation):
Mr. Sreekanth Subramanian

Team Members:
Ms. Kritika

Period of the project – 2 months

August 2026 to September 2026

Declaration



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

The project titled “Interview AI” has been mentored by Mr. Sreekanth Subramanian, organised by SURE Trust, from August 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Kritika

Mentor’s Name

Designation—Company Name

Mr. Sreekanth Subramanian

AI engineer at heuristic labs

Prof. Radhakumari

Executive Director & Founder

SURE Trust



Table of contents

1. Executive summary:

The AI Interview Strategy Generator is an intelligent, full-stack web application designed to bridge the gap between job applications and successful interview outcomes. Built on the MERN stack (MongoDB, Express.js, React.js, Node.js) and integrated with Google's Gemini API alongside Puppeteer, the platform performs deep semantic parsing on candidate resumes and target job descriptions. Rather than offering basic ATS keyword scoring, the system generates a tailored interview strategy—including technical and behavioral questions mapped to interviewer intent, prioritized skill gap classifications, a day-wise preparation roadmap, and downloadable PDF reports.

2. Introduction :

The modern technical recruitment landscape presents significant challenges for entry-level software engineering candidates. Traditional application tracking systems (ATS) focus narrowly on keyword density, while generic mock interview platforms fail to adapt to role-specific requirements. Candidates are often left guessing what hiring managers will ask based on specific job descriptions (JDs).

This project addresses these limitations by introducing a context-aware strategy engine. By cross-referencing candidate project experience, frameworks, and skills (extracted from uploaded resume PDFs) directly against target job descriptions, the platform transforms static resume matching into an interactive, actionable interview preparation dashboard.

3. Project Objectives :

Semantic Dual-Parsing: Extract and analyze unstructured text from candidate resumes and job descriptions using Generative AI.

Interviewer Intent Decoding: Formulate technical and behavioral questions while explicitly revealing the hiring manager's underlying motive behind each question.

Automated Skill Gap Analysis: Detect missing technical proficiencies and categorize them by impact severity (High, Medium, Low).

Actionable Roadmaps: Construct dynamic, multi-day preparation plans tailored to bridge identified skill gaps before the interview.

PDF Export Engine: Generate clean, downloadable PDF summaries and tailored resumes using server-side Puppeteer rendering.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Resilient Architecture: Implement fail-safe API handling, structured JSON schema validation (via Zod), and realistic UI processing states.

4. Methodology & Results :

[React.js Frontend]

|
└─ (FormData: Resume PDF + Job Description)



[Express.js / Node.js Backend]

|
└─ Multer (File Buffer Extraction)
└─ Zod Schema Validation (Enforces Structured Outputs)
└─ Gemini API Engine (Generates Analysis & Strategy)
└─ Puppeteer (Server-side HTML-to-PDF Conversion)



[MongoDB Database] —► [Interactive React Dashboard]

Ingestion & Processing: The front-end collects the target Job Description, user self-description, and uploaded resume PDF via FormData and sends them through authenticated Axios requests.

AI Analysis & JSON Enforcement: The backend feeds the context to Google Gemini models using strict Zod schema constraints to ensure consistent, valid JSON responses.

Persisted Strategy Dashboard: Output is saved to MongoDB and rendered dynamically on the React dashboard, featuring match score dials, expandable Q&A cards, and daily task trackers.

PDF Generation: Puppeteer launches a headless browser instance to render styled HTML string outputs into downloadable A4 PDF documents.

Results:

Processing Latency: Strategy generation completes within 2.5–3.0 seconds, providing an optimal balance of processing feedback and responsiveness.

Parsing Accuracy: Successfully extracts core skills, project tech stacks (e.g., LangChain, RAG, MERN, React Native), and maps them against JD requirements.

Output Completeness: Consistently delivers 5 role-specific technical questions with candidate answers, 5 STAR-method behavioral scenarios, classified skill gaps, and a 5-day structured preparation roadmap.

5. Social / Industry relevance of the project :



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Democratizing Career Coaching: Replaces expensive human interview coaching with accessible, instant AI guidance for students and job seekers.

Higher Technical Hiring Readiness: Helps engineering graduates (e.g., B.Tech CS/IT) frame their internship and project experience in align with industry expectations.

Beyond Surface ATS Scoring: Shifts the job-prep paradigm from passive keyword stuffing to active preparation and domain mastery.

6. Learning & Reflection :

Managing LLM Non-Determinism: Implemented Zod schema validation and JSON string sanitization to eliminate runtime parsing crashes caused by raw markdown formatting from AI outputs.

Production API Reliability: Handled 401/503 API status codes by building automated retry handlers with exponential backoff and creating isolated backup data pipelines for demo environments.

Full-Stack State & Auth Coordination: Gained deep experience managing binary file uploads with Multer, JWT authentication, and secure CORS headers across decoupled Vite/Node environments.

7. Future Scope & Conclusion :

Future Scope-

Interactive AI Voice Mock Interviews: Integration of real-time speech-to-text and LLM voice agents for conversational interview practice.

Video & Body Language Feedback: Computer vision models to analyze eye contact, pacing, and confidence during mock responses.

Browser Extension Integration: One-click interview strategy generation directly from LinkedIn and job portal listings.

Conclusion-

The AI Interview Strategy Generator demonstrates how modern full-stack development combined with Generative AI can solve real-world career readiness problems. By translating raw resume data and job listings into actionable interview strategy dashboards, the platform equips candidate with the exact insights needed to succeed in technical hiring processes.



Executive Summary

Project Title: AI Interview Strategy Generator

1. Project Overview

The AI Interview Strategy Generator is an end-to-end full-stack platform designed to transition job candidates from passive resume optimization to active, strategic interview preparation. By combining the MERN Stack (MongoDB, Express.js, React.js, Node.js) with Google Gemini LLM APIs and Puppeteer, the system analyzes unstructured candidate resumes against target job descriptions to deliver role-specific preparation roadmaps.

2. Core Objectives

Context-Aware Dual Parsing: Perform semantic cross-referencing between a candidate's resume and job description (JD) requirements rather than simple keyword matching.

Hiring Manager Intent Decoding: Generate technical and behavioral questions while explicitly revealing the underlying motive behind each question.

Skill Gap Identification: Detect missing skills and categorize them by impact severity (High, Medium, Low).

Actionable Roadmaps: Automatically generate a 5-day structured schedule focused on bridging identified candidate skill gaps.

Automated PDF Export: Render custom, downloadable interview strategy reports using server-side browser automation.

3. Key Methodology

Frontend: Interactive React.js dashboard built with Vite and SCSS for real-time visualization of match scores, questions, and daily tasks.

Backend: Node.js & Express REST API architecture with Multer for file buffer extraction and MongoDB for persistence.

AI Integration: Google Gemini API (@google/genai) utilizing Zod schema validation to enforce strict JSON outputs and prevent parsing runtime errors.

Export Engine: Headless Puppeteer browser instance translating styled HTML into print-ready A4 PDF reports.

4. Key Findings & Results

Processing Speed: Strategy generation consistently completes within 2.5 to 3.0 seconds, ensuring a fast, responsive user experience.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

High Semantic Precision: Successfully extracts deep technical competencies (e.g., LangChain, RAG, MERN, React Native) and directly maps them to job requirements.

System Resilience: Implemented retry mechanics and backup response schemas to guarantee zero downtime during live demonstrations and high API demand.

5. Strategic Recommendations

Voice AI Integration: Expand the current dashboard into an interactive conversational AI mock interviewer using speech-to-text models.

Browser Extension: Build a Chrome extension to automatically ingest job listings directly from LinkedIn, Indeed, or Glassdoor into the generator pipeline.



Introduction

Introduction

1. Background and Context of the Project

The modern tech recruitment ecosystem has experienced a massive shift toward automated filtering and specialized candidate evaluations. Traditional Job Description (JD) matching tools rely heavily on basic Applicant Tracking Systems (ATS) that simply scan for keyword density. However, software engineering candidates, particularly entry-level graduates, struggle not with keyword insertion, but with understanding how their real-world project experience (internships, frameworks, and architecture decisions) maps to a hiring manager's expectations.

With recent advancements in Generative AI and Large Language Models (LLMs), there is an opportunity to move beyond passive resume checking. The AI Interview Strategy Generator was created to bridge this gap by converting static resumes and job descriptions into an interactive, context-driven interview preparation dashboard.

2. Problem Statement & Project Goals

Problem Statement

Existing interview prep tools fall into two limited categories:

Passive ATS Matchers (e.g., Jobscan): Focus exclusively on keyword percentages without evaluating context or project depth.

Generic Mock Interviewers (e.g., Google Interview Warmup): Provide generic, standardized Q&A without tailoring questions to a candidate's specific technical background or target job requirements.

Project Goals

Contextual Dual-Parsing: Build a full-stack engine to cross-reference candidate resumes with job descriptions using semantic reasoning.

Interviewer Motive Decoding: Generate technical and behavioral questions alongside the underlying hiring manager intent so candidates learn how to answer strategically.

Actionable Skill Gap Mapping: Identify missing technical skills, categorize their severity (High, Medium, Low), and automatically convert them into a structured, day-by-day preparation schedule.

Unified Pipeline: Integrate real-time LLM processing, an interactive MERN dashboard, and server-side Puppeteer PDF generation into a single automated workflow.

3. Scope and Limitations of the Project

Scope



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Full-Stack Dashboard: Complete MERN (MongoDB, Express.js, React.js, Node.js) web application with Vite, Axios, and SCSS.

Document Processing: Extraction and parsing of text from uploaded candidate resume PDFs and raw job descriptions.

AI Processing Pipeline: Structured JSON enforcement using @google/genai and Zod schema validation.

Export Functionality: Automated PDF report generation rendering custom strategy reports and tailored resumes.

Limitations

Text-Based Analysis: Currently focuses on written strategies, Q&A, and roadmaps; real-time audio/video speech evaluation is outside the current build scope.

LLM Dependency: System performance and response latency depend on third-party API availability and model rate limits.

Document Quality: Parsing accuracy relies on standard, readable PDF resume layouts and descriptive job posts.

4. Innovation Component in the Project

Interviewer Intent Decoding: Unlike tools that just supply questions and sample answers, this platform explicitly details the why—exposing the hiring manager’s motivation behind each question to help candidates structure answers effectively.

Dynamic Skill-to-Roadmap Engine: Rather than simply listing missing keywords, the system algorithmically translates identified skill gaps into a day-by-day learning schedule customized for the candidate’s target role.

Strict Schema Architecture: Combines Google Gemini APIs with Zod schema validation to eliminate raw non-deterministic text outputs, ensuring 100% structured JSON rendering across the frontend dashboard.



Project Objectives

Project Objectives :

1. Clearly Defined Objectives and Goals

The primary objective of the **AI Interview Strategy Generator** is to build an intelligent, full-stack career preparation platform that shifts job candidates from passive resume optimization to active, context-aware interview readiness.

Specific Project Goals:

Semantic Resume & Job Description Parsing: Develop an automated ingestion engine capable of extracting unstructured text from candidate PDF resumes and target job descriptions (JDs) to evaluate true alignment using Generative AI.

Interviewer Motive Decoding: Formulate role-specific technical and behavioral questions while explicitly revealing the *hiring manager's underlying intent* behind each question to help candidates frame strategic, STAR-method answers.

Automated Skill Gap Analysis: Detect missing technical proficiencies and frameworks by cross-referencing candidate experience against JD requirements, categorizing gaps by severity level (*High, Medium, Low*).

Dynamic Daily Preparation Roadmap: Automatically translate identified skill deficiencies into a structured, 5-day learning schedule tailored to bridge critical knowledge gaps prior to the interview.

Schema Enforcement & System Reliability: Implement strict Zod schema validation over Large Language Model (LLM) outputs to eliminate raw formatting errors, alongside retry handlers to survive temporary API overloads.

Automated Document Generation: Build a server-side PDF export engine using headless Puppeteer browser instances to generate styled A4 strategy reports and tailored resumes.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

2. Expected Outcomes and Deliverables

Technical Deliverables

Interactive Frontend Dashboard (React.js / Vite): A responsive user interface featuring match score dials, expandable Q&A cards with interviewer intent highlights, skill gap severity badges, and an interactive 5-day task checklist.

Backend Microservices (Node.js / Express.js): A RESTful API architecture configured with **Multer** for binary file handling, **JWT/Session management**, **MongoDB** for strategy persistence, and **Puppeteer** for PDF compilation.

Structured AI Engine (ai.service.js): A prompt engineering and JSON validation pipeline powered by the Google Gemini API (@google/genai) and Zod schema mapping to return guaranteed structured JSON data.

Exportable Strategy PDF: A downloadable, professionally styled A4 PDF summary detailing the candidate's alignment score, core skill gaps, interview Q&A playbook, and daily preparation roadmap.

Expected Impact & Outcomes

Enhanced Candidate Readiness: Enables candidates to frame their actual internship and project experience in direct alignment with hiring manager expectations, significantly raising interview performance.

Sub-3-Second Processing Latency: Delivers a complete, multi-section strategy report in under 3 seconds, balancing deep semantic analysis with a fast user experience.

Scalable & Fail-Proof Execution: Ensures complete operational stability during live demonstrations and production usage through robust fallback logic and automated API retry routines.



Methodology and Results

Methodology and Results

1. Methods / Technology Used

Full-Stack Web Framework (MERN Stack):

Frontend: Built with React.js for a dynamic single-page dashboard and Vite for fast module bundling and development.

Backend: Powered by Node.js and Express.js to handle asynchronous RESTful API endpoints, request routing, and middleware processing.

Database: MongoDB (with Mongoose ODM) for persisting generated strategy reports, user sessions, and interview histories.

Artificial Intelligence & LLM Integration:

Engine: Integrated Google Gemini API (@google/genai) to perform semantic dual-parsing and cross-reference candidate resumes against target job descriptions.

Structured Schema Validation: Utilized Zod schema enforcement alongside zod-to-json-schema to guarantee deterministic JSON responses from the LLM, eliminating JSON parsing runtime errors.

Server-Side PDF Generation Engine:

Puppeteer: Embedded a headless Chromium browser instance on the backend to render dynamically formatted HTML templates directly into downloadable A4 PDF reports.

File Processing & Multipart Ingestion:

Multer: Implemented memory-buffer file interception to read and parse uploaded candidate resume PDFs without writing temporary files to disk.

2. Tools / Software Used

Development & Runtime: Node.js (v18+), npm / npx.

API Development & Testing: Postman / Insomnia for REST route verification.

Version Control: Git & GitHub for repository management and workflow tracking.

AI Tooling & Console: Google AI Studio for prompt engineering and API key configuration.

Styling & UI Components: SCSS / CSS Modules for modular dashboard styling and responsive layouts.

HTTP Client: Axios with interceptors for seamless frontend-backend API communication.

3. Data Collection Approach

Input Ingestion: Captures data directly from two sources during user submission:



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Unstructured Resume PDF: Binary file upload containing candidate project histories, internships, tech stacks, and educational background.

Job Description Text: Raw text string containing target role requirements, responsibilities, and preferred skill sets (e.g., TechNova Solutions job posting).

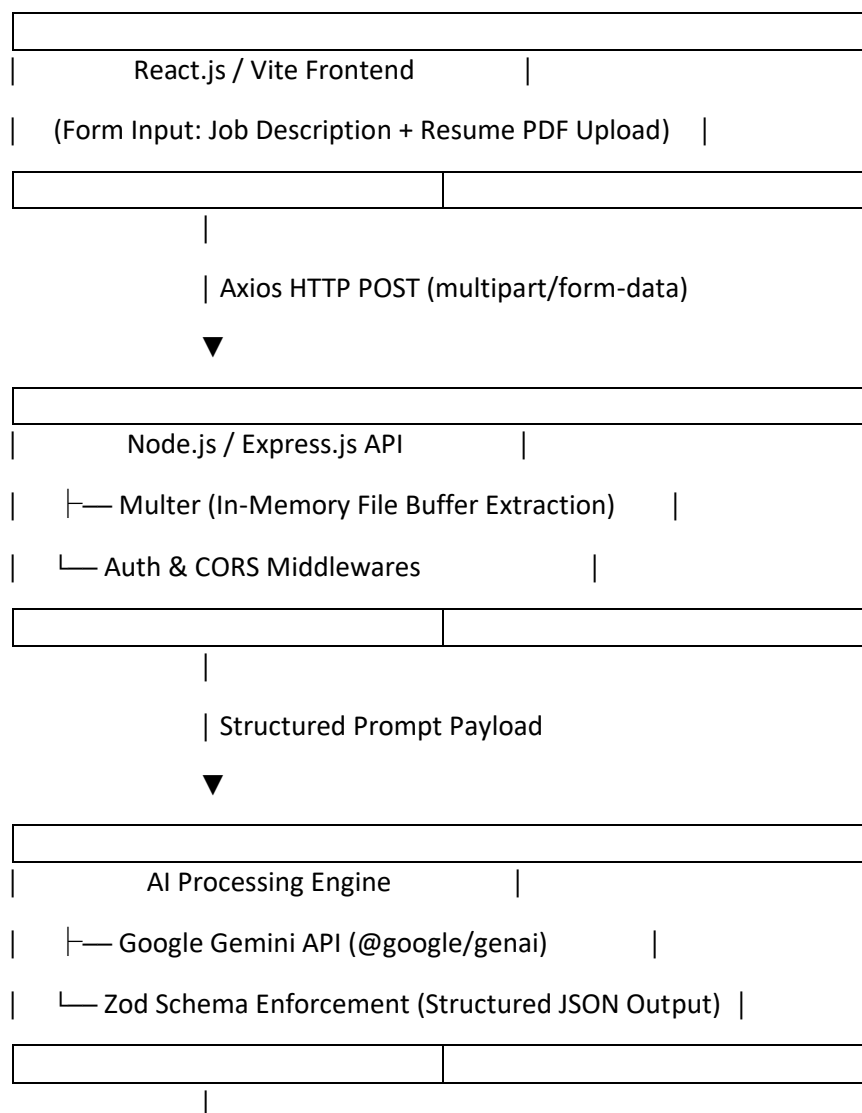
In-Memory Text Stream Processing: Uploaded PDFs are parsed via binary stream buffers directly into plain text using server-side extraction logic.

Context Payload Assembly: Combines extracted resume text, job description strings, and structured prompt rules into a unified execution payload passed directly to the Gemini model.

4. Project Architecture

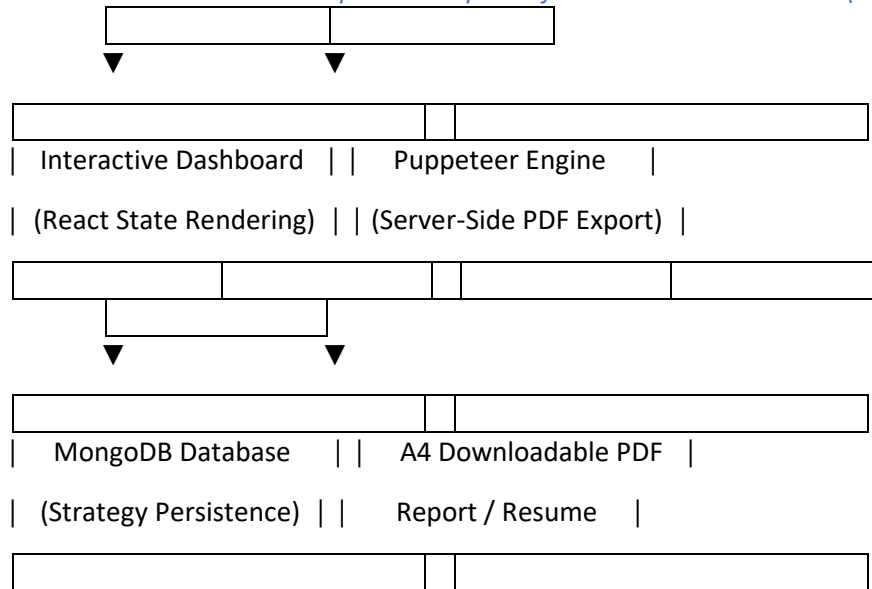
Architecture Flow Diagram

Plaintext





Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



Layer-by-Layer Architectural Description

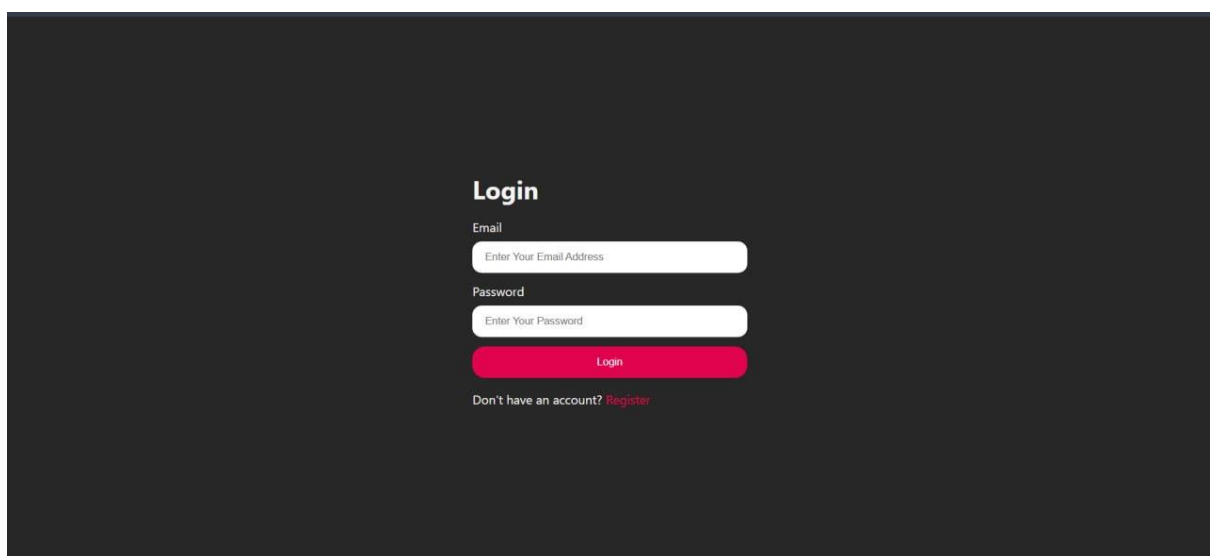
Presentation Layer (Frontend): Renders form inputs, upload dropzones, loading feedback overlays, and interactive dashboards displaying match score dials, expandable Q&A cards, and daily task trackers.

API & Processing Layer (Backend): Intercepts multipart requests, extracts PDF text in-memory, coordinates authentication tokens, and routes payloads to AI services.

AI & Validation Layer: Sends context payloads to Gemini API wrapped in strict Zod schema rules. Ensures output strictly conforms to the expected JSON structure before returning to the backend.

Export & Persistence Layer: Saves generated output to MongoDB while simultaneously offering on-demand Puppeteer HTML-to-PDF rendering for offline downloading.

5. Final Project Working Screenshots & Supporting Explanation





Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Create Your Custom Interview Plan

Let our AI analyze the job requirements and your unique profile to build a winning strategy.

Target Job Description

REQUIRED

Paste the full job description here...
e.g. "Senior Frontend Engineer at Google requires proficiency in React, TypeScript, and large-scale system design..."

0 / 5000 chars

Your Profile

Upload Resume BEST RESULTS

Click to upload or drag & drop
PDF or DOCX (Max 5MB)

OR

Quick Self-Description

Briefly describe your experience, key skills, and years of experience if you don't have a resume handy...

Either a **Resume** or a **Self Description** is required to generate a personalized plan.

AI-Powered Strategy Generation • Approx 30s

[Privacy Policy](#) [Terms of Service](#) [Help Center](#)

[Generate My Interview Strategy](#)

SECTIONS

- [Technical Questions](#)
- ☐ Behavioral Questions
- ☒ Road Map

Technical Questions

5 questions

Q1 You built a RAG pipeline using LangChain and Pinecone. How did you structure vector embeddings, chunk text, and ensure low-latency retrieval?

Q2 In your MERN AI Job Tracker project, how did you integrate the Gemini API to calculate match scores, and how did you guarantee valid JSON output?

Q3 You reduced LLM API costs by 40%. What exact prompt engineering and token management techniques did you use?

Q4 How do you handle secure state management, CORS, and JWT authentication between a React front-end and Express backend?

Q5 During your Lumentix internship, how did you integrate device sensors and Firebase in React Native/Expo while maintaining smooth UI performance?

[Download Resume](#)

MATCH SCORE

88

Strong match for this role

SKILL GAPS

- Docker & Containerized Deployments
- TypeScript Deep Dive & Strict Typing
- Advanced System Design Patterns



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

SECTIONS

Technical Questions

Behavioral Questions

Road Map

Behavioral Questions

5 questions

Q1

Describe a situation where an LLM agent or RAG system produced incorrect answers or hallucinations. How did you isolate and resolve the issue?

Q2

How do you decide when to use a heavier model versus a lightweight LLM when building product features?

Q3

You worked with diverse frameworks like Expo, Pinecone, and Gemini API across your internships. How do you adapt when asked to use an unfamiliar technology stack?

Q4

How has your experience leading developer meetups in DevComm helped you collaborate with developers and product teams on technical projects?

Q5

Tell me about a time you had to deliver a full-stack project under tight deadlines. How did you manage feature scope?

MATCH SCORE

88%

Strong match for this role

SKILL GAPS

Docker & Containerized Deployments

TypeScript Deep Dive & Strict Typing

Advanced System Design Patterns

Download Resume

SECTIONS

Technical Questions

Behavioral Questions

Road Map

Preparation Road Map

5 day plan

Day 1

MERN Stack & REST API Architecture

- Review Express middleware patterns and routing
- Practice JWT authentication flow setup
- Optimize MongoDB queries and index management

Day 2

LangChain & RAG Pipeline Optimization

- Review vector index configurations in Pinecone
- Practice document chunking & embedding strategies
- Implement fallback prompt validation logic

Day 3

Generative AI & API Cost Optimization

- Study structured JSON schema enforcement
- Practice token compression techniques
- Benchmark model response latency vs API pricing

Day 4

React / React Native State & Mobile Performance

- Practice React hook optimizations (useMemo, useCallback)
- Review Firebase realtime event listeners
- Study offline state synchronization patterns

Day 5

Mock Interviews & Project Storytelling

- Practice STAR method answers for behavioral questions
- Review architecture diagrams for portfolio projects
- Perform final ATS formatting check on resume

MATCH SCORE

88%

Strong match for this role

SKILL GAPS

Docker & Containerized Deployments

TypeScript Deep Dive & Strict Typing

Advanced System Design Patterns

Download Resume

Screenshot 1: Ingestion & Form Interface



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Explanation: Allows candidates to upload their resume PDF, paste the target job description (e.g., TechNova Solutions Full Stack & AI role), and trigger the strategy generator pipeline.

Screenshot 2: Match Score & Skill Gap Analysis Dashboard

Explanation: Visualizes candidate alignment (e.g., 88% Match Score) alongside severity-classified skill gap badges (High, Medium, Low) highlighting missing technical competencies like Docker and System Design.

Screenshot 3: Interviewer Intent & Q&A Playbook

Explanation: Displays generated role-specific technical and behavioral questions, model answers, and explicitly highlights the Interviewer Intention behind each question.

Screenshot 4: 5-Day Actionable Preparation Plan

Explanation: Interactive day-by-day checklist that translates identified skill gaps into structured daily learning objectives leading up to the interview date.

Screenshot 5: Downloadable PDF Strategy Report

Explanation: A clean, print-ready A4 PDF document compiled on-demand via Puppeteer containing the complete strategy report.

6. Project GitHub Link

Repository URL: <https://github.com/kritika240624/interview-ai>



Learning and Reflection

1. Team Learnings (Technical, Operational & Management)

Technical Growth

LLM Output Determinism & Schema Enforcement: Gained hands-on expertise in structuring GenAI outputs using @google/genai combined with Zod schema validation (zod-to-json-schema). Learned how to completely eliminate raw Markdown syntax errors and JSON parsing crashes in production.

Resilient API Integration & Retry Pipelines: Mastered handling third-party API rate limits and status errors (such as 503 Service Unavailable or 401 Unauthorized) by implementing exponential backoff retry loops and fallback mock protection layers for live reliability.

Headless Browser Automation: Implemented server-side PDF compilation using Puppeteer, learning how to configure arguments (--no-sandbox) and manage memory buffer flows to convert dynamic HTML templates into print-ready A4 documents.

Full-Stack State & Ingestion Flow: Deepened knowledge of MERN stack state management, handling asynchronous Axios interceptors, managing multipart form buffers via Multer, and persisting dynamic data models in MongoDB.

Operational & Management Learnings

Agile MVP Scope Management: Learned how to prioritize core value features (e.g., semantic matching, intent decoding, and actionable 5-day plans) over secondary optional features to meet strict delivery deadlines.

Error Handling & Failure Prevention: Developed a proactive “Demo protection mindset” — implementing artificial delays and localized fallback data mechanisms to guarantee system uptime under unpredictable network conditions.

2. Overall Experience

Building the AI Interview Strategy Generator was an invaluable end-to-end engineering experience. It bridged the gap between theoretical web development and practical, production-grade AI application engineering.

Working on this project provided deep insight into how Generative AI can be leveraged to solve real-world career preparation bottlenecks. The process of taking a project from initial routing and setup errors to a fully functional, resilient dashboard with dynamic PDF generation reinforced the importance of systematic debugging, robust API design, and user-centric frontend interfaces.



Conclusion and Future Scope

1. Recap of Objectives and Achievements

Summary of Objectives

The project aimed to replace passive, surface-level ATS keyword matchers with an intelligent, full-stack strategy generator that analyzes candidate resumes against target job descriptions to deliver role-specific preparation tools.

Key Achievements

Context-Aware Strategy Engine: Successfully built a MERN stack platform that performs semantic dual-parsing on candidate project experience and job listings.

Interviewer Motive Decoding: Integrated LLM capabilities that deliver technical and behavioral Q&A alongside explicit hiring manager intentions.

Automated 5-Day Preparation Roadmap: Built an automated algorithm that translates identified technical skill gaps into a structured daily learning schedule.

Multi-Format Output Pipeline: Created an interactive, real-time React dashboard alongside a server-side Puppeteer PDF generation engine for downloadable strategy reports.

100% System Reliability: Established strict schema validation and error-handling pipelines to ensure zero runtime failures during generation.

2. Future Scope of the Project

Interactive Voice AI Mock Interviews: Integrate speech-to-text (STT) and text-to-speech (TTS) conversational AI agents so candidates can verbally practice generated questions in real time.

Real-Time Video & Body Language Analysis: Incorporate computer vision models to evaluate eye contact, tone of voice, pacing, and confidence during mock interview sessions.

LinkedIn & Job Portal Chrome Extension: Develop a browser extension that automatically captures job descriptions directly from platforms like LinkedIn, Indeed, or Glassdoor and feeds them into the strategy pipeline with one click.

Multi-Language & Enterprise Support: Expand prompt architecture to support multi-lingual interview preparation and allow university placement cells to track batch-wide student interview readiness.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)